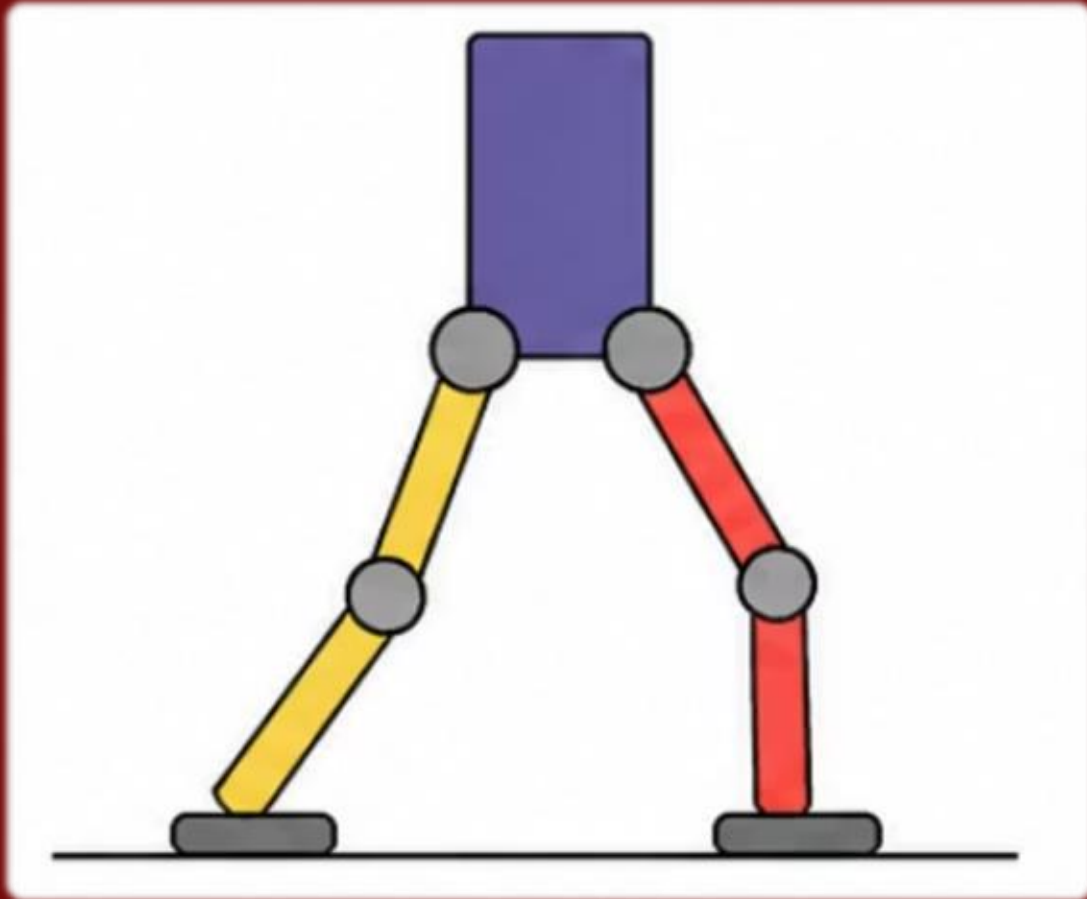




NeuroEvolution of Augmenting Topologies for Bipedal Locomotion

"Teaching a Robot to Walk Using Evolution"

-A Comparative Study of Fitness Functions, Topology Evolution, and Curriculum Learning

Presenters: Kareena Lakhani, Milan Tiwari

Course: CSE 598: Bio-inspired AI and Optimization

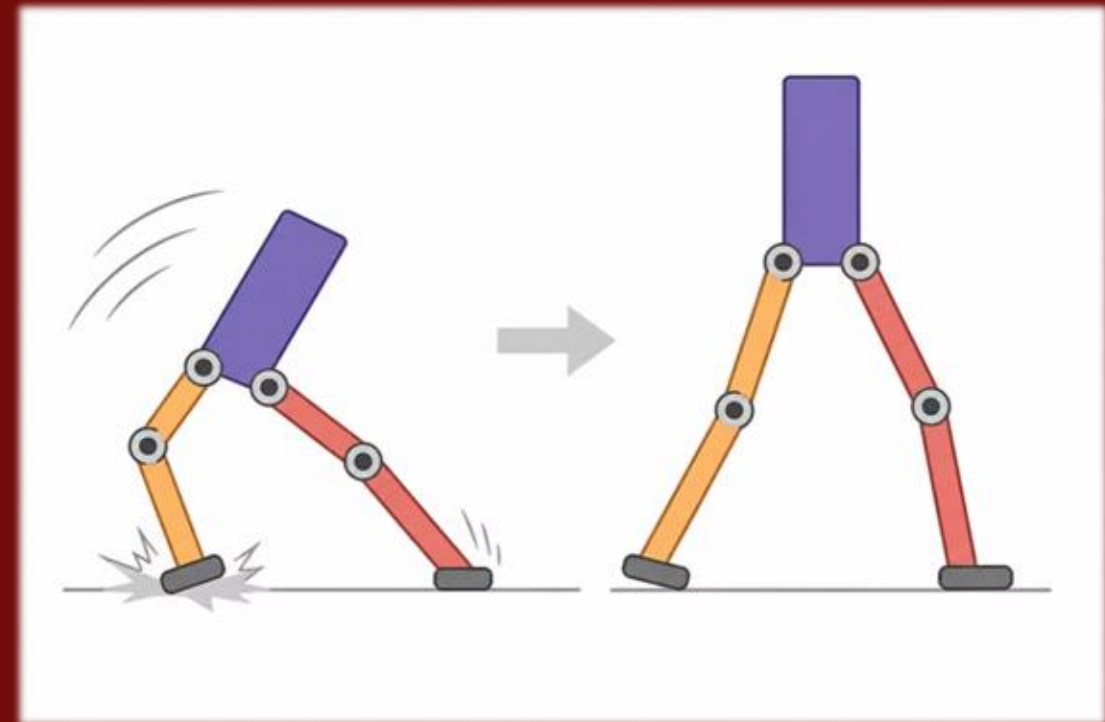
Can a Robot Learn to Walk Without Being Programmed?

The Challenge

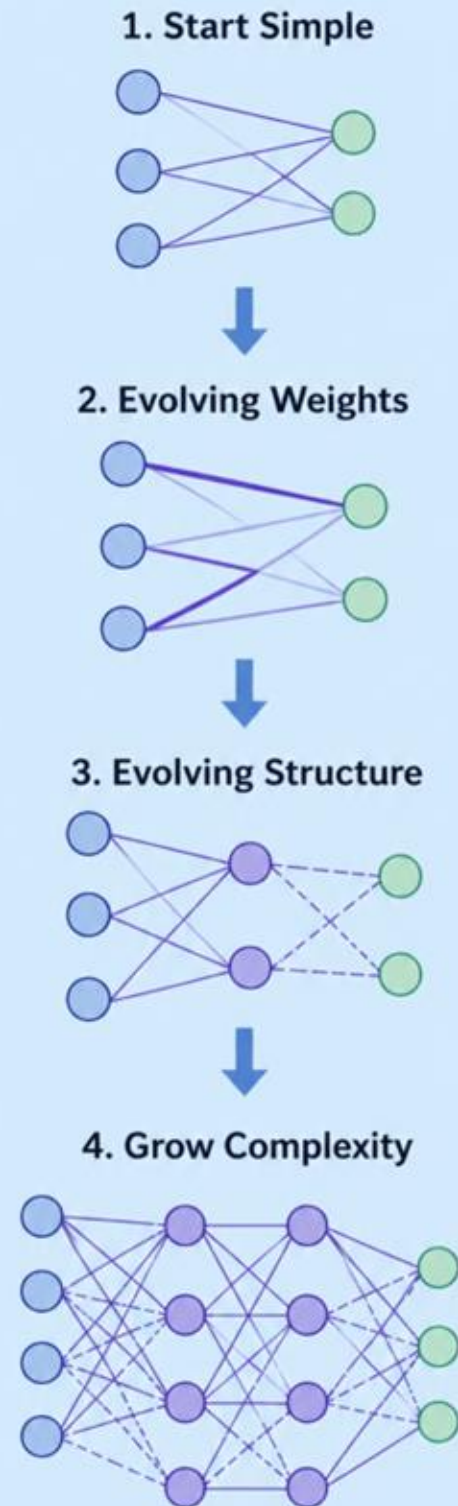
Walking is complex and hard to hand-design. Traditional approaches require predefined models and extensive programming.

The Question

Can evolution discover a walking strategy without explicit programming?



Let Evolution Design the Brain



1

Start Simple

Begin with minimal neural network structure

2

Evolving Weights

Adjust connection strengths through mutation

3

Evolving Structure

Add nodes and connections to grow complexity

4

Grow Complexity

Network architecture develops over generations

How NEAT Works



Population-Based Evolution

- NEAT starts with a *population* of single neural networks.
- Each network represents a different "candidate brain" for controlling the robot.
- Instead of training a single model, multiple networks are evaluated in parallel.



Mutation Operators

Network evolves through

Mutation:

- Weight Mutation: adjusts connection strengths
- Add connection: creates new links between neurons
- Add node: splits existing connections to grow the network.

This allows NEAT to evolve both **weights and architecture**, unlike traditional methods.



Selection and Survival

Each network is evaluated based on its performance in the environment. Better performing networks are more likely to–

- survive
- reproduce
- pass their structure to the next generation

Over time, this leads to increasingly capable behaviors.



Speciation(Key Innovation)

NEAT groups similar networks into *species*.

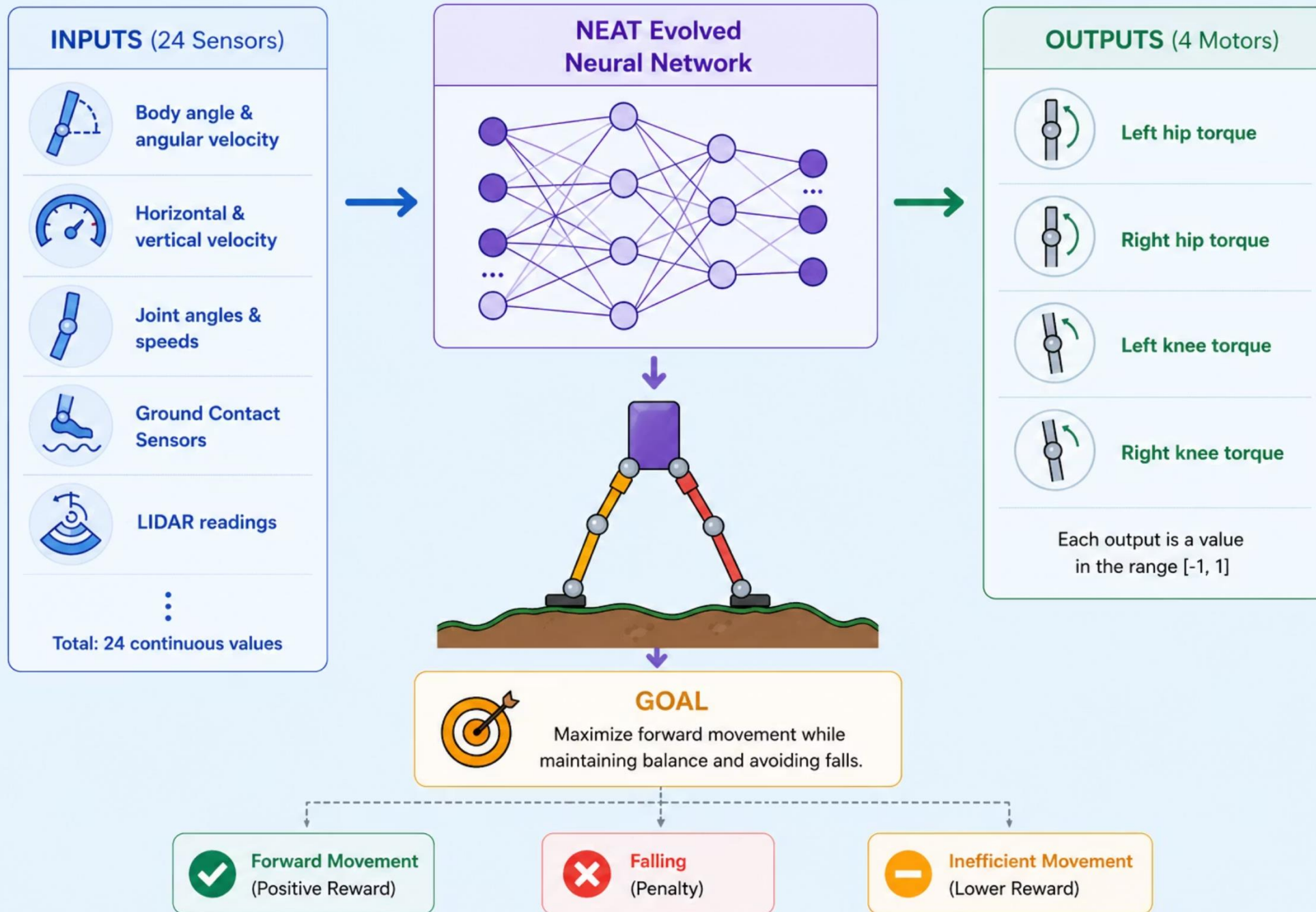
This prevents new structural changes from being eliminated too early.

Result:

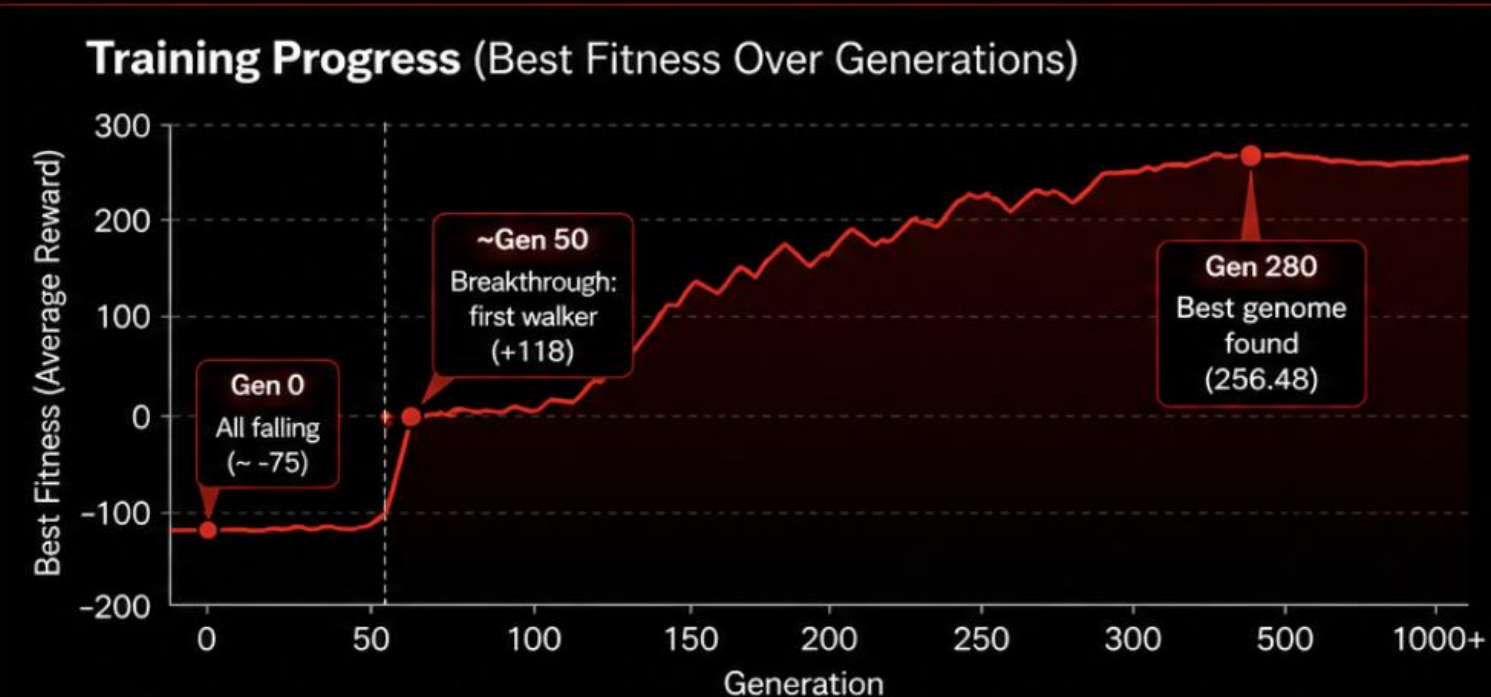
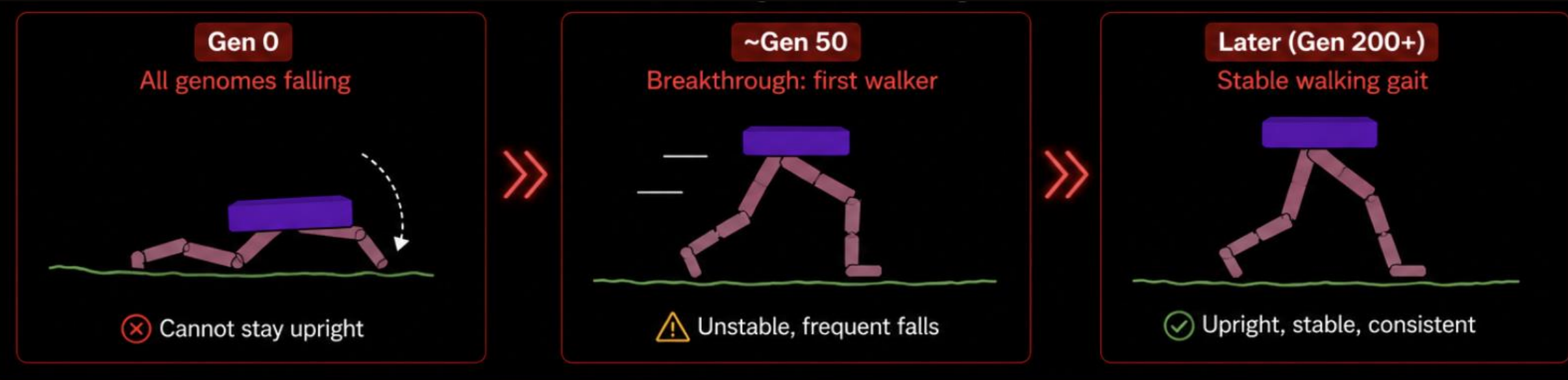
- Encourages diversity
- Allows complex structure to emerge gradually

✓ **Key Insight:** NEAT does not just learn parameters– it **evolves the structure of the network itself**.

Bipedal Walking in Physics Simulation



Evolution of Bipedal Walking



Breakthrough at Generation ~50



Best fitness jumped from **-75** to **+118** in a single generation



8 species coexisted at this point, exploring different hypotheses



Complexification at Gen 500: network grew from 13 to 34 nodes



Key Takeaway: Learning is not gradual — it is a breakthrough. Escaping local optima requires exploration and structural innovation.

Local Optimum Problem

The agent gets stuck in “standing still” because it is safe, but walking badly is heavily penalized



Safer to do nothing than fail



Result: Gets stuck in a suboptimal behavior (local optimum).



Key Insight: To solve the task, evolution must risk short-term penalties to discover better, long-term behaviors.

Does Topology Matter?

Ablation Study: Structure vs Fixed Topology

Full NEAT (Structure + Weights)

- Evolves architecture + weights
- Escapes local optima
- Learns faster and more stable behavior

Fixed Topology (Weights only)

- No structural changes
- Limited exploration
- Gets stuck in local optima



FULL NEAT ACHIEVES

+48.73

**PERFORMANCE
IMPROVEMENT**

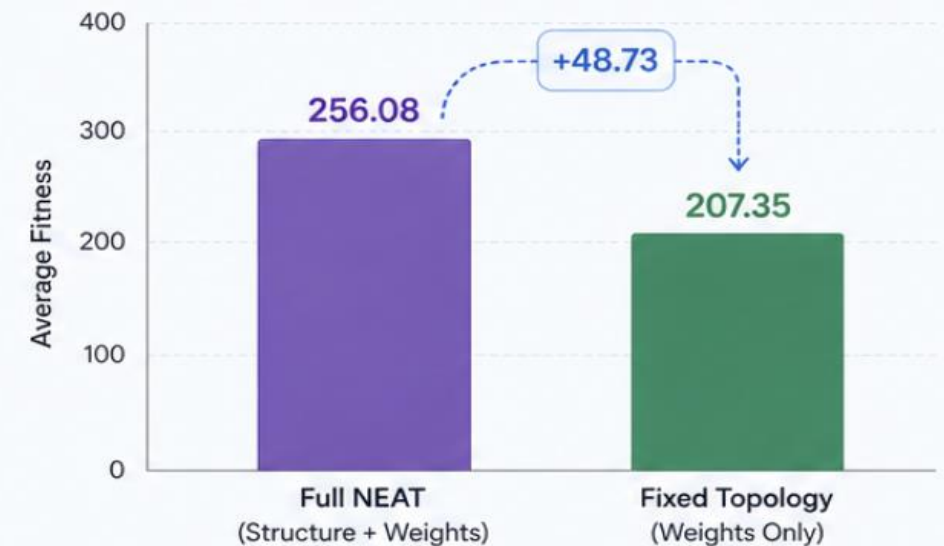
Compared to Fixed Topology

256.08 vs 207.35

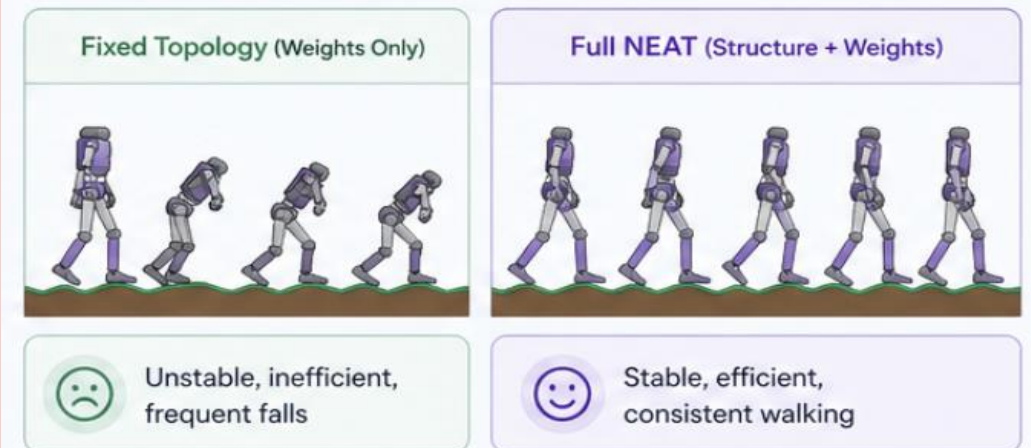
Average Fitness(Higher is better)

Performance Comparison

Average Fitness (Higher is better)



Walking Behavior Comparison

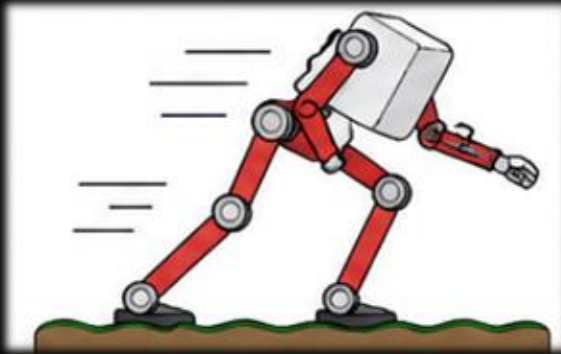


✓ **Key Insight:** Evolving network structure is essential to escape local optima and achieve superior performance

Fitness Function Shapes Behavior



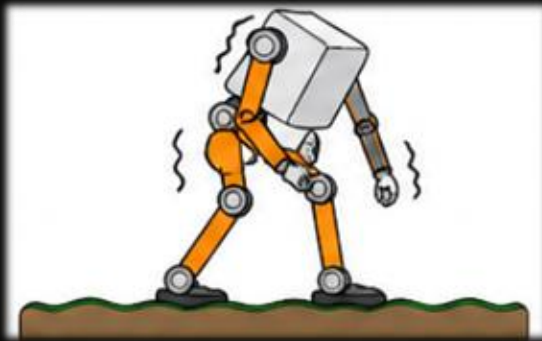
Speed Objective



- Maximizes forward movement
- Aggressive, fast gait
- Less stable, higher chance of falling



Efficiency Objective



- Penalizes energy usage
- Slow, compact movement
- Higher failure rate
- Can lead to reward hacking (standing still initially)



Stability Objective



- Penalizes tilt and imbalance
- Produces upright, natural walking
- Smooth, stable gait
- 0% failure rate

Performance Comparison



Highest Reward
272.92



0%
Failure Rate



Most Consistent
Lowest Variance

✓ **Key Insight:** Small changes in the reward function completely changes the learned behavior– stability produced the most reliable and natural walking

Stability Objective Performs Best

Rewarding stability leads to most reliable and natural walking

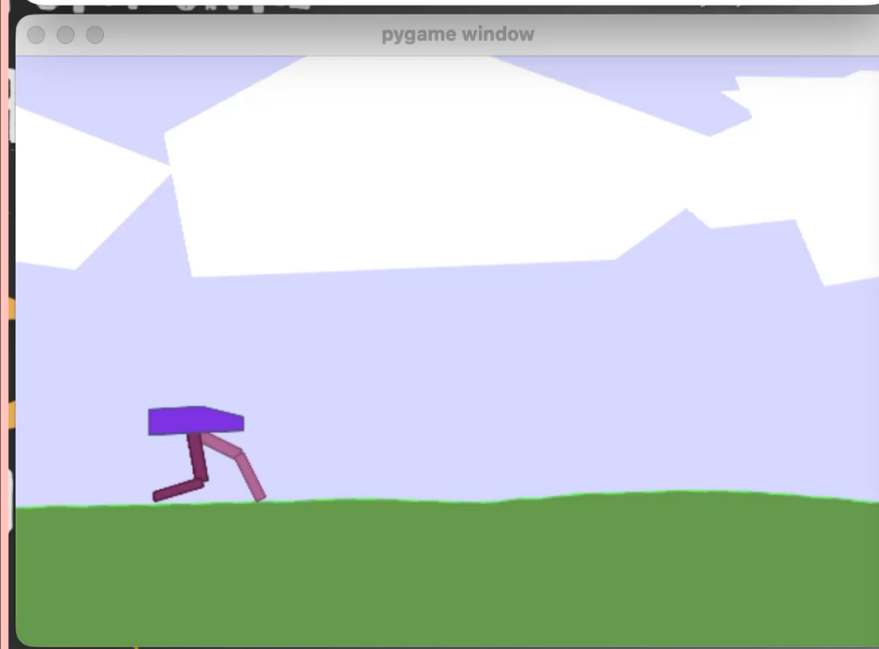
Full NEAT Walker

```
kareenalakhani — python neat_bipedal.py --mode watch — 82x24

Last login: Wed Apr 29 10:54:06 on ttys002
[(base) kareenalakhani@Kareenas-MacBook-Air ~ % cd ~/Desktop/neat_bipedal_project &
& source neat_env/bin/activate && python neat_bipedal.py --mode watch 2>/dev/null

Winner genome stats:
  Fitness:      256.48
  Nodes:        13
  Connections: 103

Watching winner walk... (close the window to stop)
Episode ended at step 1373 | Total reward: 249.13
Episode ended at step 1321 | Total reward: 253.49
Episode ended at step 1337 | Total reward: 251.69
Episode ended at step 1325 | Total reward: 253.33
```



Stability Walker

```
neat_bipedal_project — python watch_experiment.py --version stability —...

Episode ended | Steps: 1283 | Reward: 273.71
Episode ended | Steps: 1314 | Reward: 271.81
Episode ended | Steps: 1244 | Reward: 276.53
Episode ended | Steps: 1219 | Reward: 277.89
Episode ended | Steps: 1284 | Reward: 273.81
Episode ended | Steps: 1223 | Reward: 278.50
Episode ended | Steps: 1332 | Reward: 270.49
^C
Stopped.
[(neat_env) (base) kareenalakhani@Kareenas-MacBook-Air neat_bipedal_project % python
n watch_experiment.py --version stability

STABILITY winner:
  Fitness:      273.8932
  Nodes:        19
  Connections: 129

Watching stability walker... (Ctrl+C to stop)

Episode ended | Steps: 1301 | Reward: 273.37
Episode ended | Steps: 1252 | Reward: 276.44
Episode ended | Steps: 1269 | Reward: 274.67
Episode ended | Steps: 1379 | Reward: 267.94
```



HIGHEST PERFORMANCE

272.92

AVERAGE REWARD

0% Failure Rate

Most reliable performance

Highest Reward

Best overall performance

Lowest Variance

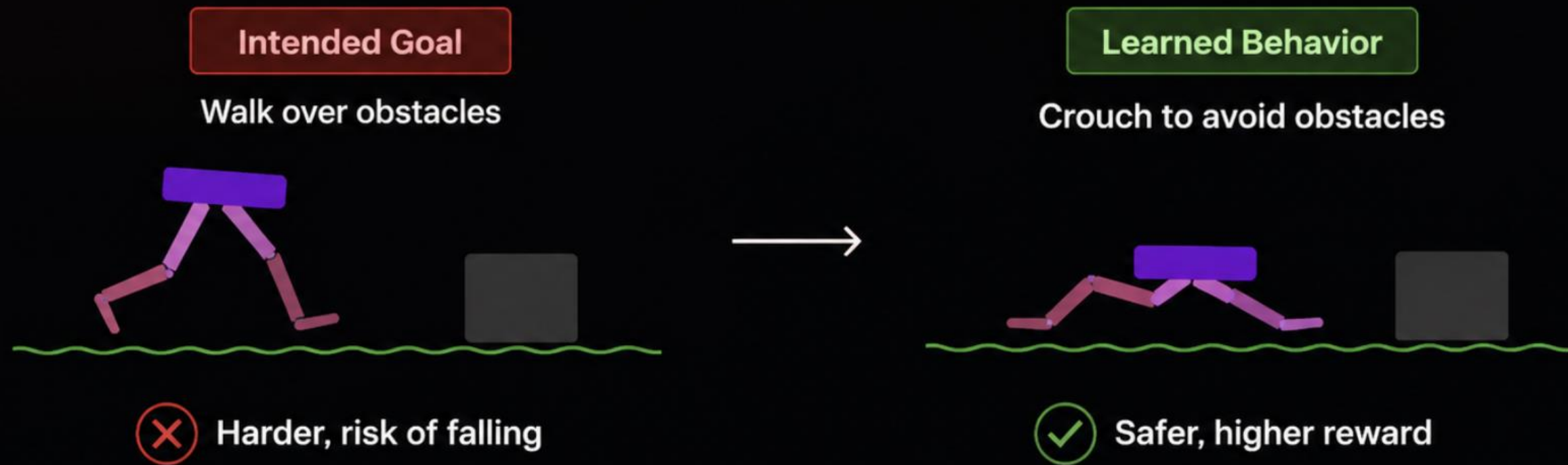
Most consistent behavior

✓ **Key Insight:** A small penalty on tilt and imbalance leads to dramatically better walking—stable, natural and highly reliable

Evolution Finds Unexpected Solutions

The agent discovers a loophole: crouching to avoid falling instead of walking over obstacles

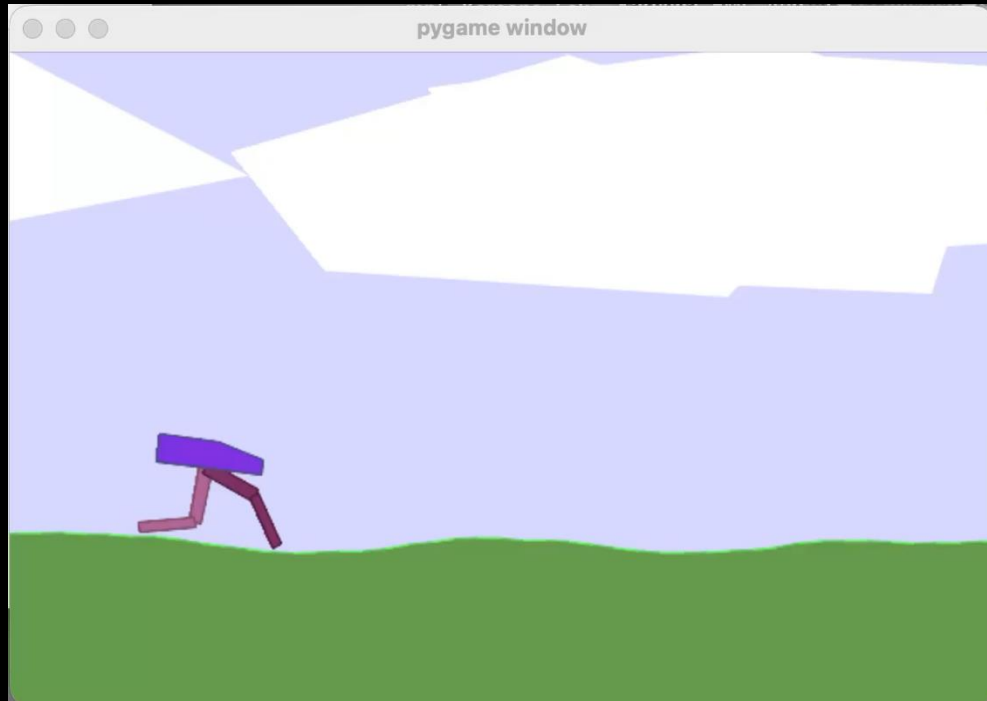
The agent optimizes the reward function, not the real goal



Key Insight: The agent optimizes the reward function, not the real goal

Generalization Failure

SEEN DURING TRAINING Flat terrain



High Performance

Average Reward: 272.92

0% failure rate

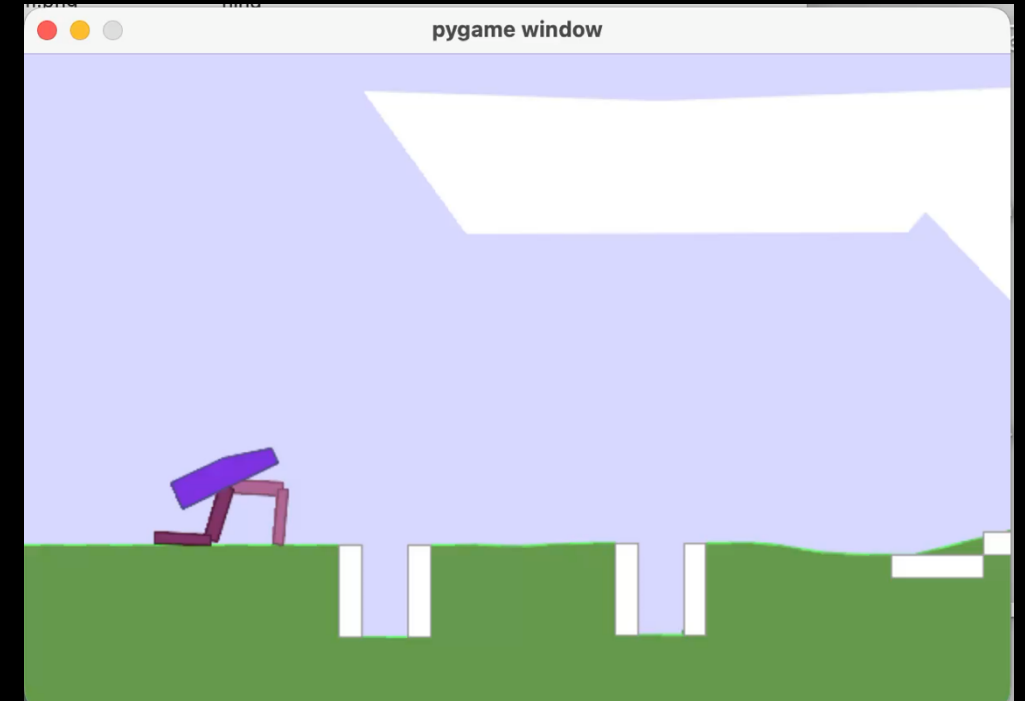
Stable and Consistent Walking

Learns effective gait

Smooth, Natural Locomotion

Performance
drops
significantly
on
unseen
terrains

UNSEEN DURING TRAINING Hardcore terrain



Poor Performance

Average Reward: 17.34

High failure rate

Often Walks and Cannot Recover

Cannot Adapt to New Conditions

No Generalization to Uneven Terrain

✓ **Key Insight:** The agent learns a solution specific to the training environment. It does not Generalize to unseen, more complex terrains.

Key Takeaways

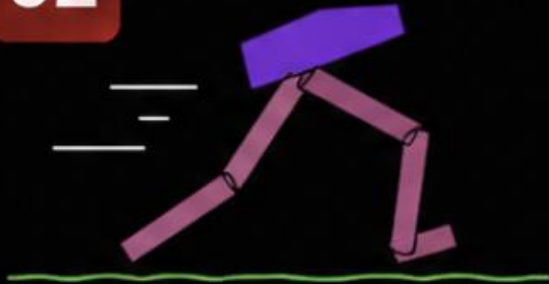
01



Structure Matters

Evolving both network structure and weights outperforms fixed topology by **+48.73** average reward.

02



Reward Defines Behavior

Different fitness functions lead to completely different walking styles and outcomes.

03



Evolution Finds Loopholes

The agent learns to crouch and avoid falling instead of walking over obstacles.



Optimizes reward, not intended behavior.

04



Fails to Generalize

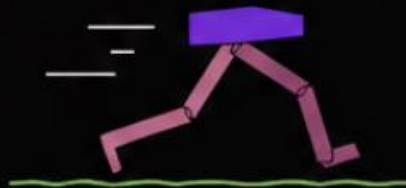
Policies learned on flat terrain perform poorly on harder, unseen terrains.

Conclusion

METHODS WE TRIED



Speed Objective



- Maximizes forward speed
- Aggressive, fast gait
- Less stable, falls often

Average Reward

242.08



Efficiency Objective



- Penalizes energy usage
- Slow, compact movement
- Higher failure rate
- Leads to reward hacking (standing still)

Average Reward

181.87



Stability Objective



- Penalizes tilt and imbalance
- Produces upright, natural walking
- Smooth and consistent gait
- 0% failure rate

Average Reward

272.92



BEST RESULT

Stability Objective



Achieved the highest reward with

272.92



Highest
Reward



0%
Failure Rate



Most Consistent
Behavior



Key Takeaway:

Among all methods, the **Stability Objective** consistently produced the most reliable and natural walking behavior with the best overall performance.